

Haoyang Zheng

PH.D. CANDIDATE AT PURDUE UNIVERSITY | MACHINE LEARNING & AI RESEARCHER

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Education

Purdue University, College of Engineering

Ph.D. in Mechanical Engineering (advised by Prof. Guang Lin, GPA 4.0/4.0)

West Lafayette, IN

Jun. 2021 - Present

Purdue University, College of Engineering

M.S. in Mechanical Engineering (GPA 3.8/4.0)

West Lafayette, IN

Sep. 2019 - May. 2021

Southwest University, School of Computer and Information Science

B.Eng. in Automation (advised by Prof. Yong Deng, Rank 1/92)

Chongqing, China

Sep. 2014 - Jun. 2018

Professional Experience

Argonne National Laboratory, Mathematics and Computer Science Division

Lemont, IL

Givens Associate

May. 2023 - Jul. 2023

- Developed scalable **Proximal Policy Optimization (PPO)** algorithms integrated with Bayesian optimization for fast policy tuning in high-dimensional control tasks.
- Developed AI-enhanced modeling and simulation pipelines for physical systems by combining machine learning with PDE solvers.

Research Interests

I study **Generative Modeling** and **Reinforcement Learning** (RL) from a probabilistic modeling perspective. My interests include **LLMs**, continuous and discrete diffusion, RL, and sampling/optimization methods that improve latency and robustness. My recent work on **discrete diffusion distillation** achieves over **60×** faster language generation than ARMs with similar quality. My goal is to build fast and reliable LLMs.

Selected Papers

- [1] **Haoyang Zheng**, Xinyang Liu, Xiangrui Kong, Nan Jiang, Zheyuan Hu, Weijian Luo*, Wei Deng*, Guang Lin, “**Ultra-Fast Language Generation via Discrete Diffusion Divergence Instruct**”, *arXiv:2509.25035* (2025).
- [2] Weixin Wang, **Haoyang Zheng**, Guang Lin, Wei Deng, Pan Xu*, “**Rethinking Langevin Thompson Sampling from A Stochastic Approximation Perspective**”, *NeurIPS 2025 Workshop on Dynamics at the Frontiers of Optimization, Sampling, and Games (DynaFront)*.
- [3] **Haoyang Zheng**, Ruqi Zhang*, Guang Lin*, “**Exploring Non-Convex Discrete Energy Landscapes: A Langevin-Like Sampler with Replica Exchange**”, *arXiv:2501.17323* (2025).
- [4] **Haoyang Zheng**, Hengrong Du, Qi Feng, Wei Deng*, Guang Lin*, “**Constrained Exploration via Reflected Replica Exchange Stochastic Gradient Langevin Dynamics**”, *ICML 2024*.
- [5] **Haoyang Zheng**, Wei Deng*, Christian Moya, Guang Lin*, “**Accelerating Approximate Thompson Sampling with Underdamped Langevin Monte Carlo**”, *AISTATS 2024*.
- [6] **Haoyang Zheng**, Yao Huang, Ziyang Huang, Wenrui Hao, Guang Lin*, “**HomPINNs: Homotopy Physics-Informed Neural Networks for Inverse Problems**”, *Journal of Computational Physics* (2024).
- [7] **Haoyang Zheng**, Jeffrey R. Petrella, P. Murali Doraiswamy, Guang Lin*, Wenrui Hao, “**Data-Driven Causal Model Discovery and Personalized Prediction in Alzheimer’s Disease**”, *Nature Digital Medicine* (2022).

Talks

- [1] On Sampling Tasks with Langevin Dynamics, Brown University, July 2024.
Host: George Em Karniadakis.

Posters

- [1] Reflected Replica Exchange Stochastic Gradient Langevin Dynamics, September 2024, on [Mathematical and Scientific Foundations of Deep Learning Annual Meeting \(Simons Foundation\)](#), at New York, NY.
- [2] Accelerating Thompson Sampling with Underdamped Langevin Monte Carlo, October 2024, on [SIAM Conference on Mathematics of Data Science](#), at Atlanta, GA.
- [3] On Thompson Sampling with Underdamped Langevin Monte Carlo, September 2023, on [Mathematical and Scientific Foundations of Deep Learning Annual Meeting \(Simons Foundation\)](#), at New York, NY.

Achievements

SELECTED AWARDS

2024	Student Travel Awards, 2024 Mathematical and Scientific Foundations of Deep Learning Annual Meeting	New York, NY
2024	Student Travel Awards, 2024 SIAM Conference on Mathematics of Data Science (MDS24)	Atlanta, GA
2024	Student Travel Awards, Nonlocality: Challenges in Modeling and Simulation	Providence, RI
2023	Student Travel Awards, 2023 Mathematical and Scientific Foundations of Deep Learning Annual Meeting	New York, NY
2018	Pacemaker to Technological Innovation in Chongqing, Awarded 10 college students every two years	Chongqing, China
2018	Outstanding undergraduates in Chongqing, Awarded 1% of all college undergraduate students every year	Chongqing, China
2015	China National Scholarship, Awarded 1% of all undergraduates in China	China

SELECTED HONORS

2017	Finalist (0.5%), Interdisciplinary Contest in Modeling	China
2016	Special prize (2/3568), International Mathematical Contest in Modeling	China

Service

Conference Reviewer	ICLR, ICML, NeurIPS, AISTATS, AAAI SIAM Multiscale Modeling and Simulation, Journal of Computing and Information Science in Engineering,
Journal Reviewer	Transactions on Machine Learning Research, Journal of Supercomputing, Scientific Reports, Knowledge and Information Systems

Skills & Tools

Machine Learning	Generative Modeling, Reinforcement Learning, Bayesian Optimization, Sampling
ML/DL Frameworks	PyTorch, DDP, OpenAI Gym, HuggingFace
Programming Languages	Python, Bash, MATLAB
Uncertainty Quantification	Bayesian Inference, Langevin Dynamics, MCMC